ROLL NO: | | | | | | | | | | |

(To fill the roll no is compulsory)

EVEN SEMESTER EXAMINATION 2024-25

B.Tech. Cloud Computing

Semester: VI

SUBJECT CODE: BCSICT602

ETHICAL HACKING

Time: 03:00 Hrs. **Maximum Marks: 70**

NOTE: THE QUESTION PAPER CONTAINS THREE SECTIONS. ATTEMPT ALL THE SECTIONS.

SECTION – A

NOTE: Attempt all parts of the following questions. (14 Marks)

Q. 1. Answer the following questions: **(1×7 = 7 Marks)**

(i) What is Cyber Ethical Hacking?
(ii) Define a Cyber Attack.
(iii) What are Trojans in cyber security?
(iv) What is Network Scanning used for?
(v) What is DNS Spoofing?
(vi) What is Malware Threat in cyber security?
(vii) What is the significance of Firewalls in ethical hacking?

Q. 2. Answer the following questions: **(1×7 = 7 Marks)**

(i) What is Social Engineering?
(ii) What is a Denial of Service (DoS) attack?
(iii) Explain Session Hijacking.
(iv) What is the purpose of Hacking Web Servers?
(v) What is IoT Hacking?
(vi) What is Ethical Hacking in Cloud Computing?
(vii) How can Social Engineering be used in hacking?

SECTION – B

NOTE:Attempt all questions. **(7×2×2 = 28 Marks)**

Q.3. (i)(a) Explain the concept of Cyber Attacks. Discuss different types of Cyber Attacks with real-world examples.

OR

(b) Describe the various phases of hacking in detail. How does each phase contributes to an ethical hacker's approach?

(ii)(a) What is Information Gathering? Explain the different techniques used in footprinting and reconnaissance.

OR

(b) Discuss the role of scanning in ethical hacking. What are the common network scanning tools and their functionalities?

Q.4. (i)(a) Define malware. Explain the different types of malwares, including viruses, worms, Trojans, and backdoors.

OR

(b) Explain social engineering attacks in detail. What are the different techniques used in social engineering?

(ii)(a) Describe the process of system hacking. What are the different Ways attackers exploit vulnerabilities to gain unauthorized access?

OR

(b) Explain malware threats in detail. How do malware attacks affect organizations, and what measures can be taken to prevent them?

SECTION – C

NOTE: Attempt all questions. **(14×2 = 28 Marks)**

Q. 5.(a) Explain the concept of session hijacking. How do attackers hijack active sessions, and what are the countermeasures?

OR

(b) Explain the role of cryptography in cyber security. How does encryption help in protecting sensitive data?

Q. 6. (a) Describe different types of Denial-of-Service (DoS) attacks and their impact on business operations.

OR

(b) Explain the role of malware in cyber-attacks. How do malware threats evolve, and what techniques are used for detection and prevention?